

Module 01 - Authentication & User Identity

Firebase Authentication + user profile bootstrap

SOURCE OF TRUTH: This resource intentionally uses plain JavaScript. The locked stack is React + JavaScript + Firebase + Material UI (MUI). Do not introduce Next.js, TypeScript, shadcn/ui, Redux, or another backend framework unless the project owner explicitly changes this specification.

Scope

Own the sign-up, login, logout, password reset, Google sign-in, protected routes, session persistence, and creation of the corresponding users/{uid} Firestore profile.

Routes

```
/login
/signup
/forgot-password
/app/* (protected)
```

Firebase behavior

- Use Firebase Authentication for credentials and provider identity.
- After first successful authentication, ensure users/{uid} exists. If missing, create it using Auth profile data plus safe defaults.
- Use onAuthStateChanged (or equivalent React hook) as the single source of auth session truth.
- Logout clears local session-dependent state; do not manually store passwords or auth tokens in Firestore.

User document

```
users/{uid}
{
  displayName,
  email,
  photoURL,
  createdAt,
  updatedAt,
  lastLoginAt,
  onboardingComplete
}
```

Validation

- Email must be valid.
- Password rules must be explicit in the UI and match Firebase configuration.
- Prevent duplicate profile creation by using uid as the document key.
- Handle invalid credentials, disabled accounts, rate limits, popup cancellation, and network errors.

UI states

State	Required behavior
Initial auth load	Centered MUI CircularProgress or skeleton; no flash of protected screen
Empty login	Clear fields + validation
Submitting	Disable submit + progress indicator

State	Required behavior
Success	Navigate to dashboard
Error	Inline field/form message; no raw Firebase error dump

Acceptance criteria

- New user can sign up and lands in the app.
- Existing user can log in after refresh.
- Google user is also provisioned in users/{uid}.
- Password reset produces a clear success/error state.
- Protected routes redirect unauthenticated users to /login.
- Authenticated users cannot see login page content except by explicit logout.

Non-goals

- Phone OTP
- Enterprise SSO
- Role management outside group membership
- Custom JWT backend